Online PAC Labeling from a Linear Programming Perspective

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Recap: AI-assisted labeling

We have a pool of unlabeled items $X_1, \ldots, X_T$. Need to collect their true labels $Y_1, \ldots, Y_T$.

Two ways to get a label $\tilde{Y}_t$:

- **ML models** $\widehat{Y}_t^{(a)}$ — sometimes *wrong*, but *cheap*
- **Expert** Y_t — *correct*, but *expensive*.

Two Goals:

- Save cost by reducing the number of expert queries.
- Probably Approximately Correct (PAC) guarantee:

$$\frac{1}{T}\sum_{t=1}^{T} \ell(Y_t, \widetilde{Y}_t) \ \leq \ \varepsilon \quad w.p. \geq 1 - \alpha.$$

Problem setup

- For each item X_t, decide how to label it: use the **expert** or one of k **cheap models**.
- Choose action a_t from an action set $\mathcal{A} = \{E, 1, \ldots, k\}$.
- For item t and action a, a labeling **loss** and a **cost**:

$$h_t(a) = \begin{cases} 0, & a = E \\ \ell(Y_t, \widehat{Y}_t^{(a)}), & a \in [k] \end{cases} \qquad c_t(a) = \begin{cases} 1, & a = E \\ 0, & a \in [k] \end{cases}$$

- Label the item with $\widetilde{Y}_t = Y_t$ if $a_t = E$, else $\widehat{Y}_t^{(a_t)}$.

Key quantities: *cost* we want to minimize, *error* we must control.

The cost-aware labeling LP

Let $x_{t,a} \in [0,1]$ be the probability of taking action a on item t.

$$\min_x \quad \underbrace{\frac{1}{T} \sum_t \sum_a c_t(a)\, x_{t,a}}_{\text{average cost}}$$

$$\text{s.t.} \quad \underbrace{\frac{1}{T} \sum_t \sum_a h_t(a)\, x_{t,a} \le \varepsilon,}_{\text{average error budget}} \qquad \sum_a x_{t,a} = 1, \quad x_{t,a} \ge 0.$$

- **One** global constraint: keep average error $\le \varepsilon$.
- Naturally handles *multiple* cheap sources and item-specific costs.
- If true losses $h_t(a)$ were known, this LP is the *optimal* policy.

The dual gives a single "price of error"

One Lagrange multiplier $\lambda \geq 0$ for the error constraint. The dual:

$$\max_{\lambda \geq 0} \; -\lambda\varepsilon + \frac{1}{T}\sum_{t=1}^{T} \min_{a \in \mathcal{A}} \left\{ c_t(a) + \lambda\, h_t(a) \right\}.$$

The fixed-price rule

Once λ is fixed, the decision **separates across items**:

$$a_t(\lambda) \in \arg\min_{a \in \mathcal{A}} \left\{ c_t(a) + \lambda\, h_t(a) \right\}.$$

λ is the exchange rate between cost and error.

Small $\lambda \Rightarrow$ trust AI | Large $\lambda \Rightarrow$ defer to expert.

What the price actually does: a threshold

With equal cheap-source costs, first **route** to the best cheap model

$$j_t^\star \in \arg\min_{j\in[k]} \widehat{h}_t(j), \qquad s_t = \min_j \widehat{h}_t(j) \quad \text{(the ``uncertainty score'')}.$$

Then λ only decides *accept vs. defer* via a threshold $1/\lambda$:

$$\boxed{s_t \geq 1/\lambda \;\Rightarrow\; \text{defer to expert}} \qquad s_t < 1/\lambda \;\Rightarrow\; \text{accept cheap label.}$$

- Recovers the original **PAC-labeling** threshold rule (single source: $\widehat{h}_t = \text{uncertainty}$).
- Multi-source extension is "free": just replace the score by $\min_j \widehat{h}_t(j)$.
- Two regimes ahead: **offline** (fix λ once) and **online** (move λ_t over time).

Part I — Offline PAC Labeling

Certify one price on a calibration set, then label the whole pool.

Offline: certify a price from a small labeled sample

Idea. The cost $C_T(\lambda)$ is known from proxies; only the error $L_T(\lambda)$ needs true labels. Estimate it on a calibration set.

- Sample m items, get expert labels, form empirical loss $\widehat{L}_m(\lambda)$.
- Hoeffding upper confidence bound:

$$U_m(\lambda; \alpha) = \widehat{L}_m(\lambda) + \sqrt{\frac{\log(1/\alpha)}{2m}}.$$

- Pick the **cheapest certified price**:

$$\hat{\lambda} = \min\{\lambda \geq 0 : \; U_m(\lambda; \alpha) \leq \varepsilon\}.$$

Why no union bound over λ?

As λ grows, items only move *AI* $\rightarrow$ *expert*. So $L_T(\lambda)$ is **monotone** — a single nested family, certified for free.

Offline: the PAC guarantee

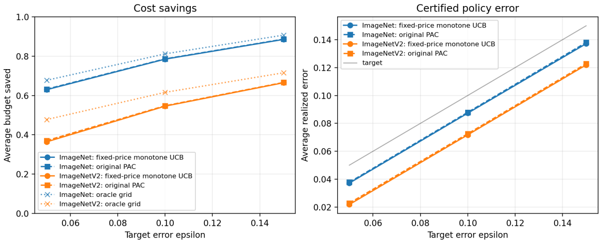
Theorem (offline monotone-threshold PAC)

With probability at least $1-\alpha$ over the calibration draw, the released labels satisfy

$$\frac{1}{T}\sum_{t=1}^{T}\ell(Y_t, \widetilde{Y}_t) \;\le\; \varepsilon.$$

- Distribution-free, finite-sample. Falls back to expert-only if no finite price qualifies.
- **Corollary:** on a fresh deployment pool of size n, error $\le \varepsilon + \sqrt{\log(1/\delta)/2n}$ w.p. $1-\alpha-\delta$.
- Proxies affect *cost*, not *validity*: a bad proxy just defers more often.

Offline experiments: LP rule matches original PAC



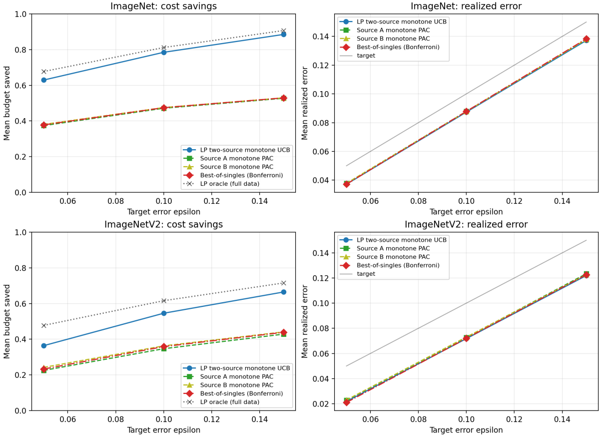
One cheap source, ImageNet / ImageNetV2.

One source ($\varepsilon = 0.10$):

Data	Method	Error	Saved
IN	LP	0.087	0.785
IN	PAC	0.088	0.786
INv2	LP	0.072	0.546
INv2	PAC	0.073	0.548

- Error stays under target ε (0 violations).
- LP fixed-price $\approx$ original PAC, as predicted.
- Saves $\sim$78% of expert budget on ImageNet.

Offline experiments: routing beats single sources



Two complementary cheap sources (by class parity).

Two sources ($\varepsilon = 0.10$, ImageNet):

Method	Error	Saved
LP routing	0.087	**0.785**
Best-of-singles	0.088	0.475

- Per-item routing to the lower-proxy source.
- Same error control, but **much** larger budget savings (0.78 vs 0.47).
- This is where the LP view earns its keep.

Part II — Online PAC Labeling via Audits

Items stream in; labels are hidden unless we audit.

Online PAC: moving threshold with randomized audits

Items arrive one at a time. An AI label's true loss is *hidden* unless we pay the expert.

Maintain a **moving price** λ_t, i.e. a threshold $1/\lambda_t$. Each round, route to the best model ($s_t = \min_j \widehat{h}_t(j)$), then:

- $s_t \geq 1/\lambda_t$: **defer** $\rightarrow$ expert label (correct).
- $s_t < 1/\lambda_t$: release the AI label, then **audit** it with probability p.

Audit

With probability p: query the expert, *correct* the released label, and *reveal* the AI's true loss ℓ_t.

Audits buy occasional, randomized feedback — just enough to steer the price.

The price update: learn from audited feedback

Dual ascent on the price

$$\lambda_{t+1} = \big[\, \lambda_t + \eta\, \widehat{Z}_t \,\big]_+, \qquad \widehat{Z}_t = \begin{cases} -\varepsilon, & \text{deferred to expert,} \\ (\ell_t - \varepsilon)/p, & \text{AI label } \textit{audited}, \\ 0, & \text{AI label, not audited.} \end{cases}$$

- Audited **mistake** $(\ell_t > \varepsilon) \Rightarrow \widehat{Z}_t > 0 \Rightarrow \lambda \uparrow \Rightarrow$ threshold $1/\lambda \downarrow \Rightarrow$ defer more.
- Deferral or clean audit $\Rightarrow \lambda \downarrow \Rightarrow$ trust the AI more.
- The $1/p$ re-weighting makes rare audits count correctly: on average the update sees the true AI error, $\mathbb{E}[\widehat{Z}_t \mid \text{past}] = (\text{AI error before audit}) - \varepsilon$.

η: step size | ε: target error (the update centers on it directly).

Finite-horizon guarantee

Theorem (audited online control)

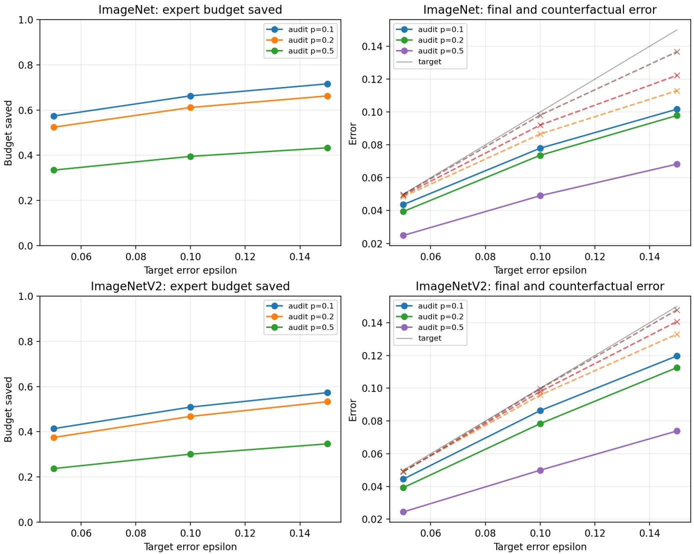
For *any* data sequence, with probability $\geq 1-\delta$ over the audit draws,

$$\frac{1}{T}\sum_{t=1}^{T}\ell(Y_t,\widetilde{Y}_t) \;\leq\; \varepsilon + \underbrace{\frac{\Lambda_{\mathrm{on}}}{\eta T}}_{\substack{\text{optimization}\\ O(1/T)}} + \underbrace{\frac{1}{p_{\min}}\sqrt{\frac{2\log(1/\delta)}{T}}}_{\substack{\text{partial feedback}\\ O(1/\sqrt{T})}} \;=\; \varepsilon + B_T.$$

- $\Lambda_{\mathrm{on}}/(\eta T)$: how far the price can drift; vanishes like $1/T$.
- $\frac{1}{p_{\min}}\sqrt{\cdots}$: noise from auditing only a fraction; smaller audit rate $\Rightarrow$ larger.
- $B_T = O(T^{-1/2}) \to 0$, so the released error approaches ε as the stream grows.

assume proxy floor ($\widehat{h}_t \geq \rho$), $\Lambda_{\mathrm{on}} = \max\{\lambda_1,\ 1/\rho + \eta(1-\varepsilon)/p_{\min}\}$ caps the price: once $\lambda_t \geq 1/\rho$, the threshold $1/\lambda_t \leq \rho$ drops below proxy floor, so *every* item defers, and λ_t can no longer increase.

Online experiments



Online ($\varepsilon = 0.10$):

Data	p	Error	Saved
IN	0.1	0.078	0.663
IN	0.2	0.074	0.612
IN	0.5	0.049	0.395
INv2	0.1	0.086	0.509
INv2	0.2	0.078	0.468
INv2	0.5	0.050	0.301

- Smaller p: save more budget.
- Larger p: correct more, lower final error.
- The audit rate p trades cost vs. error

Takeaways

One idea, two regimes

- LP dual $\Rightarrow$ a single *price of error* λ.
- **Offline:** certify $\hat{\lambda}$ on held-out labels.
- **Online:** move λ_t with audited inverse-probability feedback.

What we showed

- Proxies $\widehat{h}_t$ affect *cost*, not *validity*.
- LP rule matches PAC with one source; *routing wins* with two.
- Online stays valid under partial feedback; p trades cost vs. error.

Next step

Current data has only one real cheap source. The interesting case is many sources with *heterogeneous costs*, where a scalar threshold no longer suffices.

Thank you! **Questions?**